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Correct Answer

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Practice Exam - 1

Question 59 of 117



A 60 year old patient received a continuous flow LVAD 4 years ago for Stage D non-ischemic cardiomyopathy. His device was implanted prior to routine use of outflow graft stabilization hardware (bend relief clip). He is currently admitted for volume overload and worsening heart failure. In the last 24 months he has had 4 admissions for decompensated heart failure. The left ventricle has progressively dilated during this time despite repeated increases in LVAD speed. TEEs have shown only trace aortic regurgitation. During this admission the patient is noted to have new low flow alarms. LDH is approximately 370 u/L (about 1.5 his baseline value) and RHC shows a wedge pressure of 30 mmHg and cardiac index of 1.7 L/min/m², which did not improve with an increase in LVAD pump speed of 600 RPM. A CTA is performed and shown below. The patient ultimately undergoes a LVAD pump exchange and an image of his explanted device is below.

What is the etiology of the patient's pump dysfunction?





(Figure 1)

Figure 1: CT CT Angiography



(Figure 2)

Figure 2: LVAD Pump

A. Worsening aortic insufficiency

 B. Extra-luminal outflow graft compression

C. Intraluminal thrombosis of the outflow graft

 D. Outflow graft twisting

E. Progression of underlying cardiomyopathic process

Testing Point: The Board-certified AHFTC specialist should recognize etiologies of persistently poor left ventricular unloading in patients with continuous flow LVADs.

Correct Answer: D. outflow graft twisting.

Rationale: The patient presents with worsening heart failure despite the presence of a durable LVAD. His clinical picture is highlighted by evidence of slowly progressive poor LV unloading as demonstrated by his increasingly dilated LV, high wedge pressure and resultant HF hospitalizations and eventual development of low flow alarms. While other processes may present similarly (progressive AI, chronic pump thromboses, etc), the question stem gives enough information to exclude them and determine that an outflow graft complication is most likely. The CTA image provided shows a clear high grade out flow graft twist and the exploration of the explanted device confirms that as well.

is likely to blame.

Answer A: He does not have AI on TEE.

Answer C: His LDH is not in range to raise concern for an acute pump thrombosis.

This complication has been most commonly described in modern continuous-flow LVADs, particularly HeartMate 3, in which twisting of the outflow graft can occur near the swivel/bend relief mechanism, resulting in progressive obstruction and reduced LV unloading. Although this complication has become much less common with contemporary implantation techniques and routine use of graft stabilization hardware, patients with older (“legacy”) devices remain at risk. Recognition is important because outflow graft twisting can cause progressive heart failure, low-flow alarms, and hemodynamic compromise, and treatment typically requires surgical revision, pump exchange, or transplantation.

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